GEOMETRIC IRREDUCIBILITY CAN BE CHECKED AFTER AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the converse implication; no step is referred to the reader.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-irreducible-check-after-extension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Existence of a common extension field of two extensions of a field. Fields, Lemma fields-lemma-common-extension-field, invoked at proof lines 1356-1358. Supplies the field k"' over which both base changes become comparable.
- Surjectivity is stable under base change and the spectrum of a field extension surjects onto the base. Standard, used at proof lines 1358-1361. Supplies that irreducibility passes down along X_{k"'} to X_{k"}.

Source reference. Original lemma 0H7K: sources/stacks-chow/tags.txt, source lines 21460–21460. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a scheme over the field k. Let k'/k be a field extension. Then X is geometrically irreducible over k if and only if $X_{k'}$ is geometrically irreducible over k'.*

Proof. If X is geometrically irreducible over k, then it is clear that $X_{k'}$ is geometrically irreducible over k'. For the converse, for any field extension k''/k there exists a common field extension k'''/k' and k'''/k'', see Fields, Lemma 0H7K. As the morphism $X_{k'''} \to X_{k''}$ is surjective (as a base change of a surjective morphism between spectra of fields) we see that the irreducibility of $X_{k'''}$ implies the irreducibility of $X_{k''}$. Thus if $X_{k'}$ is geometrically irreducible over k' then X is geometrically irreducible over k. □

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